Lagrange Multipliers

When the constraint is an equation to find the maximum and minimum value of $f(x,y,z)$ subject to the constraint $g(x,y,z)=k$, where k is a constant

$$\vec{\nabla} f = \lambda \vec{\nabla} g$$

Where λ (lamda) is the Lagrange Multipler

$f \longleftarrow$ Optimize
$g \longleftarrow$ constraint

example Problem

$F(x,y,z) = x^2+y^2+z^2$ 　 $x^2+y^2+z^2+xy=12$

- this is $g(x,y,z)$ we know this because it is equal to a constant

began by taking the gradient for each equation

$\vec{\nabla} f = \langle 2x, 2y, 2z \rangle$ 　 $\vec{\nabla} g = \langle 2x+y, 2y+x, 2z \rangle$

$\vec{\nabla} f = \lambda \vec{\nabla} g \implies \langle 2x, 2y, 2z \rangle = \lambda \langle 2x+y, 2y+x, 2z \rangle$

<u>equations</u>

- include the Constrian $g(x,y,z)$ in your equations

$$2x = 2x\lambda + y\lambda$$
$$2y = 2y\lambda + x\lambda$$
$$2z = 2z\lambda$$
$$x^2+y^2+z^2+xy=12$$

We can use any equation here but choose the simplist one

$$2z = 2z\lambda$$
$$2z - 2z\lambda = 0$$
$$2z(1-\lambda) = 0$$

• do Not can the 2z you will lose Potential answers Factor it out

$$1-\lambda = 0 \qquad\qquad 2z = 0$$
$$\lambda = 1 \qquad\qquad z = 0$$

$$2x = 2x(1) + y(1)$$
$$2x = 2x + y$$
$$\boxed{y = 0}$$

$$2(0) = 2(0)(1) + x(1)$$
$$\boxed{x = 0}$$

$$(0)^2 + (0)^2 + z^2 + (0)(0) = 12$$
$$\sqrt{z^2} = \sqrt{12}$$
$$\boxed{z = \pm\sqrt{12}}$$

• First two equation do Not have z only one that has z in it is the last one but if we do Plug in the information We already Know:

$$(0)^2 + (0)^2 + (0)^2 + (0)(0) = 12$$
$$0 \neq 12$$

$$2x = 2x\lambda + y\lambda$$
$$2x\frac{(1-\lambda)}{\lambda} = y$$

$$2y = 2y\lambda + x\lambda$$
$$2\left(\frac{2x(1-\lambda)}{\lambda}\right) = 2\left(\frac{2x(1-\lambda)}{\cancel{\lambda}}\right)\cancel{\lambda} + x\lambda$$
$$\frac{4\cancel{x}(1-\lambda)}{\lambda} = 4\cancel{x}(1-\lambda) + \cancel{x}\lambda$$
$$4(1-\lambda) = \lambda(4 - 4\lambda + \lambda)$$
$$4 - 4\lambda = 4\lambda - 3\lambda^2$$
$$3\lambda^2 - 8\lambda + 4 = 0$$

$$\lambda = 2 \text{ and } z = 0 \qquad\qquad \lambda = 2/3 \text{ and } z = 0$$

$\lambda = 2$ and $z = 0$

$2x = 2x(2) + (2)y$

$-\cancel{2}x = \cancel{2}y$

$x = -y$

Plug this into

$x^2 + y^2 + z^2 + xy = 12$

$(-y)^2 + y^2 + \cancel{(0)^2} - y^2 = 12$

$2y^2 - y^2 = 12$

$\sqrt{y^2} = \sqrt{12}$

$y = \pm\sqrt{12}$

$\lambda = 2/3$ and $z = 0$

$2y = 2y\left(\frac{2}{3}\right) + x\left(\frac{2}{3}\right)$

$2y = \frac{4y}{3} + \frac{2}{3}x$

$\frac{\cancel{2}y}{\cancel{3}} = \cancel{\frac{2}{3}}x$

$y = x$

Plug this into $x^2 + y^2 + z^2 + xy = 12$

$x^2 + x^2 + \cancel{(0)^2} + x^2 = 12$

$3x^2 = 12$

$\sqrt{x^2} = \sqrt{4}$

$x = \pm 2$

Points

$(0, 0, +\sqrt{12})$

$(0, 0, -\sqrt{12})$

$(\sqrt{12}, -\sqrt{12}, 0)$

$(-\sqrt{12}, \sqrt{12}, 0)$

$(2, 2, 0)$

$(-2, -2, 0)$

Use lagrange multipliers to Find the exterme Values of $f(x,y) = 2x^2+3y^2-4x-5$
Subject to the Constriants $x^2+y^2 \leq 16$

$$\vec{\nabla} f = \langle 4x-4,\ 6y \rangle \qquad \vec{\nabla} g = \langle 2x,\ 2y \rangle$$

$$\vec{\nabla} f = \lambda \vec{\nabla} g$$

* When you have a inequality Just make it a regular Function

equations

$4x-4 = 2x\lambda$

$6y = 2y\lambda$

$x^2+y^2 = 16$

$$6y = 2y\lambda$$
$$6y - 2y\lambda = 0$$
$$2y(3-\lambda) = 0$$
$$2y = 0 \qquad 3-\lambda = 0$$
$$y = 0 \qquad \lambda = 3$$

$y=0$:

$$x^2 + \cancel{(0)^2} = 16$$
$$\boxed{x = \pm 4}$$

$\lambda = 3$:

$$4x-4 = 2x\lambda$$
$$4x-4 = 6x$$
$$\boxed{x = -2}$$
$$x^2+y^2 = 16$$
$$(-2)^2+y^2 = 16$$
$$\sqrt{y^2} = \sqrt{12}$$
$$\boxed{y = \pm\sqrt{12}}$$

Points

$(-4,0)$

$(4,0)$

$(-2,\sqrt{12})$

$(-2,-\sqrt{12})$

Find the critical Point at $f_x = 0$ and $f_y = 0$

$$\left.\begin{aligned} f_x &= 4x-4 \\ f_y &= 6y \end{aligned}\right\} \quad \begin{aligned} x &= 1 \\ y &= 0 \end{aligned} \implies (1,0)$$

a @ $F(-4,0)$
b @ $F(4,0)$
c @ $F(-2,-\sqrt{12})$
d @ $F(-2,\sqrt{12})$
e @ $F(1,0)$

$f(x,y,z) = 4x+4y+3z \quad , \quad 2x^2+2y^2+3z^2 = 19$

$\vec{\nabla} f = \langle 4,4,3 \rangle \qquad \vec{\nabla} g = \langle 4x, 4y, 6z \rangle$

<u>equations</u>

$4 = 4x\lambda$
$4 = 4y\lambda$
$2x^2+2y^2+3z^2 = 19$

$3 = 6z\lambda \qquad 4 = 4y\lambda \qquad 4 = 4x\lambda$

$\frac{1}{2z} = \lambda \qquad \frac{1}{\lambda} = y \qquad \frac{1}{\lambda} = x$

$$2\left(\frac{1}{\lambda}\right)^2 + 2\left(\frac{1}{\lambda}\right)^2 + 3\left(\frac{1}{2\lambda}\right)^2 = 19$$

$$\frac{4}{\lambda^2} + \frac{3}{4\lambda^2} = 19$$

$$\frac{16+3}{4\lambda^2} = 19 \implies \lambda = \pm\frac{1}{2}$$

$4 = 4\lambda x \implies 4 = 4x(\pm 1/2) = x=2,\ x=-2$

$4 = 4y\lambda \implies 4 = 4y(\pm 1/2) = y=2,\ y=-2$

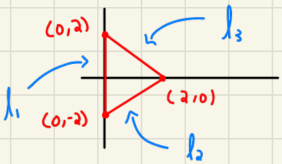
$3 = 6z\lambda \implies 3 = 6z(\pm 1/2) = z=-1,\ z=1$

<u>Points</u>

$(-2,-2,-1)$ @ $F(-2,-2,-1) = -19$
$(2,2,-1)$ @ $F(2,2,1) = 19$

$$\boxed{\begin{array}{l} \text{Max} = 19 \\ \text{Min} = -19 \end{array}}$$

$F(x,y) = x^2 + y^2 - 2x$, D is closed triangle region with vertices $(2,0)$, $(0,2)$. and $(0,-2)$



$l_1 = x = 0$ ($x=0$ stay's the same in that region)

$F(0,y) = x^2 + (0)^2 - 2x \Rightarrow y^2$

For y in $[2,-2]$ Find max/min (Plug these into the y^2)

Max $= 4$

$l_2 = \begin{array}{cc} (0,-2) & (2,0) \\ x_1 \; y_1 & x_2 \; y_2 \end{array}$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{0-(-2)}{2-0} = \frac{2}{2} = 1$$

$y = x - 2$ So the y is going to be $y = x - 2$

$F(x, x-2) = x^2 + (x-2)^2 - 2x \Rightarrow F(x,x-2) = 2x^2 - 6x + 4$

$g(x) = 2x^2 - 6x + 4$ Find max and min in region $[0,2]$

we Just called it $g(x)$ so you don't get confused

$g'(x) = 4x - 6$

$4x - 6 = 0$

$x = \frac{3}{2}$

$g(0) = 4$

$g(2) = 0$

$g(3/2) = -1/2$

Max $= 4$, Min $= -1/2$

$l_3 = (0,2) \; (2,0)$
$\quad\; x_1\, y_1 \;\; x_2\, y_2$

$m = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{(0-2)}{(2-0)} = -1 \qquad y = -x+2$

$F(x, -x+2) = x^2 + (-x+2)^2 - 2x \implies 2x^2 - 6x + 4$

$g(x) = 2x^2 - 6x + 4$ Find max and min in $[0,2]$

$g'(x) = 4x - 6$

$4x - 6 = 0$

$x = 3/2$

$g(0) = 4$

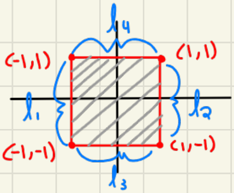
$g(2) = 0$

$g(3/2) = -1/2$

Max $= 4$ min $= -1/2$

$\boxed{\begin{array}{l} \text{abs max} = 4 \\ \text{abs min} = -1/2 \end{array}}$

$F(x,y) = x^2 + y^2 + x^2y + 4, \; D = \{(x,y) \mid |x| \le 1, |y| \le 1\}$



$l_1 = x = -1$ (the $x=-1$ never changes in l_1)

$F(-1,y) = 1 + y^2 + y + 4 \implies y^2 + y + 5$

Max $= 7$ min $= 5$

$\left.\begin{array}{l} y=-1 \\ y=0 \\ y=1 \end{array}\right\}$ Plug into y^2+y+5

$F(-1) = 5$

$F(0) = 5$

$F(1) = 7$

$l_2 = x=1$ (the $x=1$ is the Same along the l_2 line)

$F(1,y) = 1+y^2+y+4 \implies y^2+y+5$

$y=-1$, $y=0$, $y=1$ } Plug into y^2+y+5

$F(-1)=5$

$F(0)=5$

$F(1)=7$

Max $=7$ Min $=5$

$l_3 = y=-1$ (the $y=-1$ we be the same along the l_3 line)

$F(x,-1) = \cancel{x^2}+1-\cancel{x^2}+4 = 5$

$x=-1$ $F(-1)=5$

$x=0$ $F(0)=5$

$x=1$ $F(1)=5$

Max $=5$

$l_4 = y=1$ (the $y=1$ is the Same along the l_4 line)

$F(x,1) = x^2+1+x^2+4 \implies 2x^2+5$

Max $=7$ min $=5$

$x=-1$ $F(-1)=7$

$x=0$ $F(0)=5$

$x=1$ $F(1)=7$

$\boxed{\text{absolute max} = 7 \qquad \text{absolute min} = 5}$

to get the absolute max get all the max and min values you have gotten and Find the absolute max and min

Find the Shortest distance from the Point $(2,0,-3)$ to the Plane $x+y+z=1$

✱ to Find the Shortest distance From a Point to a Plane use Point to Plane distance Formula

$$\text{distance} = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

rewrite the equation:

$$x+y+z=1 \implies x+y+z-1=0$$

$A=1,\ B=1,\ C=1,\ D=-1$

Point $(x_0, y_0, z_0) = (2, 0, -3)$

$$\text{Distance} = \frac{|(1)(2) + (1)(0) + (1)(-3) - 1|}{\sqrt{1^2 + 1^2 + 1^2}} = \boxed{\frac{2}{\sqrt{3}}}$$

- double integral over rectangles

if $F(x,y)$ is continous on the rectangle $R = [(x,y) \mid a \le x \le b,\ c \le y \le d]$
then:

$$\iint_R F(x,y)\, dA = \int_a^b \int_c^d F(x,y)\, dy\, dx = \int_c^d \int_a^b F(x,y)\, dx\, dy$$

the average value of a continous function $F(x,y)$ on the rectangle $R = [(a,b) \mid a \le x \le b,$
$c \le y \le d]$ is given by $f_{average}$

$$f_{avg} = \frac{1}{\text{area of } R} \iint_R F(x,y)\, dA$$

example problems:

$$\int_1^4 \int_0^2 (x+y)^2\, dx\, dy$$

- For dx we took the integral with respect to x and treated the y as a constant then on dy we took the integral with respect to y and treated the x as a constant

$$\int_0^2 (x+y)^2\, dx$$

$u = x+y$
$du = 1dx$
$du = dx$

$$\int_0^2 (u)^2\, du$$

$$\left[\frac{(x+y)^3}{3} \right]_0^2 = \left(\frac{(2+y)^3}{3} \right)$$

$$\int_1^4 \frac{(2+y)^3}{3}\, dy \longrightarrow \frac{1}{3} \int_1^4 (2+y)^3\, dy$$

$u = 2+y$
$du = dy$

$$\frac{1}{3} \int_1^4 (u)^3\, du$$

$$\frac{1}{3} \left[\frac{(2+y)^4}{4} \right]_1^4 = \frac{1}{3} \left[\frac{(2+4)^4}{4} - \frac{(2-1)^4}{4} \right]$$

$$\boxed{\frac{1295}{12}}$$

$$\int_1^3 \int_1^5 \frac{\ln x}{xy}\, dy\, dx$$

$$\int_1^5 \frac{\ln x}{xy}\, dy = \frac{\ln x}{x} \int_1^5 \frac{1}{y}\, dy$$

$$\frac{\ln x}{x}\left[\ln(y)\right]_1^5 \longrightarrow \frac{\ln x}{x}\left(\ln(5) - \ln(1)\right)$$

$$\frac{\ln(x)\cdot \ln(y)}{x}$$

$$\int_1^3 \frac{\ln(x)}{x}\, dx$$

$u = \ln x$

$du = \frac{1}{x}\, dx$

$x\,du = dx$

$$\int_1^3 \frac{(u)}{\cancel{x}} \cdot \cancel{x}\, du$$

$$\left[\frac{(\ln(x))^2}{2}\right]_1^3 = \left[\frac{(\ln(3))^2}{2} - \frac{(\ln(1))^2}{2}\right]$$

$$\boxed{\frac{(\ln(3))^2}{2}}$$

$$\int_0^1 \int_0^2 y e^{x-y}\, dx\, dy$$

$$\int_0^2 y e^{x-y}\, dx \longrightarrow \int_0^2 y e^{x} \cdot e^{-y}\, dx$$

$$\int_0^2 y e^{-y} \cdot e^{x}\, dx \longrightarrow \left[y e^{-y} \cdot e^{x}\right]_0^2$$

$$\left[y e^{-y} \cdot e^{2}\right] - \left[y e^{-y} \cdot e^{0}\right]$$

$$y e^{-y}(e^2 - 1)$$

$$\int_0^1 y e^{-y}(e^2-1)\, dy$$

$$(e^2-1)\int_0^2 y e^{-y}\, dy$$

you need integration by parts for the rest

$$uv - \int v\, du$$

$u = y$ $\quad$ $dv = e^{-y}$

$du = dy$ $\quad$ $v = \int e^{-y} = -e^{-y}$

$$-(y)(e^{-y}) + \int e^{-y}\, dy$$

$$(e^2-1)\left[-y e^{-y} - e^{-y}\right]_0^2$$

$$(e^2-1)\left[-2e^{-2} - e^{-2}\right] - \left[-1\right]$$

$$\boxed{(-3e^{-2}+1)(e^2-1)}$$

Made with Goodnotes

evaluate the integral $\iint_A \left(\frac{x}{y+x^2}\right) dA$ where $R = [(x,y) \mid, 0 \le x \le 1,\ 2 \le y \le 3]$

$$\int_2^3 \int_0^1 \frac{x}{y+x^2}\, dx\, dy$$

$$\int_0^1 \frac{x}{y+x^2}\, dx$$

$u = y + x^2$

$du = 2x\, dx$

$\frac{du}{2x} = dx$

$$\Longrightarrow \frac{1}{2}\int_0^1 \frac{\cancel{x}}{u} \cdot \frac{du}{\cancel{x}} \qquad \left[\frac{1}{2} \ln(y+x^2)\right]_0^1$$

$$\frac{1}{2} \ln|y+1| - \frac{1}{2} \ln|y|$$

- you need to solve this by integration by parts

$uv - \int v\, du$ $\qquad$ $u =$ lipet

$$\int_2^3 \frac{1}{2}\ln|y+1| - \frac{1}{2}\ln|y|\, dy$$

$u = \ln|y+1|$ $\qquad$ $dv = dy$

$du = \frac{1}{y+1}$ $\qquad$ $v = \int dy = y$

$u = \ln y$ $\qquad$ $dv = dy$

$du = \frac{1}{y}$ $\qquad$ $v = \int dy = y$

$$y\ln|y+1| - \int \frac{y+1-1}{y+1}\, dy$$

$$y\ln|y+1| - \int \frac{\cancel{y+1}}{\cancel{y+1}} - \frac{1}{y+1}$$

$u = y+1$

$du = dy$

$$y\ln|y+1| - \int 1 - \frac{1}{y+1}\, du$$

$$\left[y\ln|y+1| - y - \ln|y+1|\right]_2^3$$

$$y\ln y - \int \cancel{y} \cdot \frac{1}{\cancel{y}}\, dy$$

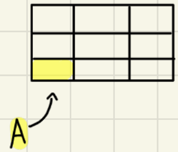
$$\left[y\ln y - y\right]_2^3$$

$$\boxed{[3\ln 4 - 3 - \ln 4] - [2\ln 3 - 2 - \ln 3] + [3\ln 3 - 3] - [2\ln 2 - 2]}$$

- Not simplified

• Riemann sum

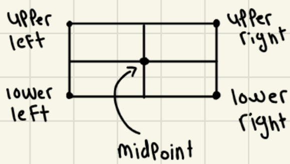
- Volume per rectangle



Where A is a area of triangle (small)

Pay attention to the details it might ask you for the upper right, upper left, lower left or lower right

this is A the small rectangle



example problem:

Point on x-axis

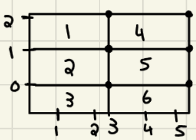
Points on y-axis

number of parts along x-axis

if $R = [1,5] \times [-1,2]$ use riemann sum with $m=2$ and $n=3$ to estimate the value of $\iint_R (1 - xy^2)\, dA$ by:

number of parts along x-axis

take the sample points of upper right corner of each square



So now they want you to sum every right corner

Volume = $F(x,y) \cdot A$

1) $F(3,2) \cdot 2 = -22$

2) $F(3,1) \cdot 2 = -4$

3) $F(3,0) \cdot 2 = 2$

4) $F(5,2) \cdot 2 = -38$

5) $F(5,1) \cdot 2 = -8$

6) $F(5,0) \cdot 2 = \cdot 2$

add them all up

$= \boxed{-68}$

finding area

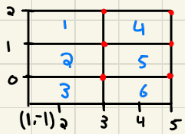
$$\Delta x = \frac{5-1}{2} \qquad \frac{[1,5] = 5-1}{\text{number of boxes on } x\text{-axis } (m)} = 2$$

$$\Delta y = \frac{2-(-1)}{3} = \frac{[-1,2] = 2-(-1)}{\text{number of boxes on } y\text{-axis } (n)} = 1$$

area = $\Delta x * \Delta y = 2$

Riemann Sum For Center or mid Point

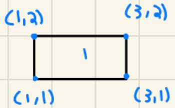
$m=2,\ n=3$



$\Delta x = \frac{5-1}{2} = 2$

$\Delta y = \frac{2-(-1)}{2} = 1$

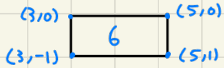
$A = \Delta x \cdot \Delta y = 2$



$x = \frac{3+1}{2} = 2$

$y = \frac{1+2}{2} = \frac{3}{2}$

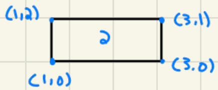
$F(2, \frac{3}{2}) \cdot 2 =$



$\Delta x = \frac{3+5}{2} = 4$

$\Delta y = \frac{-1-(-1)}{2} = 0$

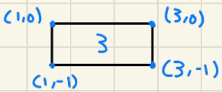
$F(4,0) \cdot 2 =$



$\Delta x = \frac{1+3}{2} = 2$

$\Delta y = \frac{0+2}{2} = 1$

$F(2,1) \cdot 2 =$



$\Delta x = \frac{1+3}{2} = 2$

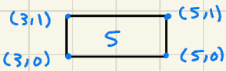
$\Delta y = \frac{-1+0}{2} = -1/2$

$F(2, -1/2) \cdot 2 =$

$\Delta x = \frac{3+5}{2} = 4$

$\Delta y = \frac{1+1}{2} = 1$

$F(4,1) \cdot 2 =$



$\Delta x = \frac{3+5}{2} = 4$

$\Delta y = \frac{0+1}{2} = 1/2$

$F(4, 1/2) \cdot 2 =$

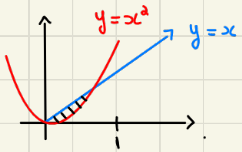
Plug those numbers into the given equation For us here we will use the Same equation From last Page the add up all the answers and thats it I did Not Finish the Problem but the rest Consists on Plugging into equation and adding

• double integral over general region

<u>example</u>

$0 \le x \le 1$

$x^2 \le y \le x$



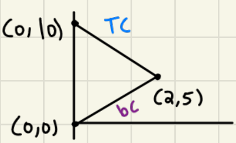
top

$$\int_0^1 \int_{x^2}^{x} f(x,y)\, dx\, dy$$

bottom

✯ when you are dealing with these types of integral you began the integral limit from the bottom and the variable always goes on the inside

• we are going from top to bottom with respect to x



bottom $\le y \le$ top

$\frac{5}{2}x \le y \le -\frac{5}{2}x + 10$

TC: $\underset{x_1 \; y_1}{(0,10)}, \underset{x_2 \; y_2}{(2,5)}$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5-10}{2-0} = -\frac{5}{2}$

y-int = $(0,10)$

$y = mx + b = -\frac{5}{2}x + 10$

bc: $\underset{x_1 \; y_1}{(0,0)}, \underset{x_2 \; y_2}{(2,5)}$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5-0}{2-0} = \frac{5}{2}$

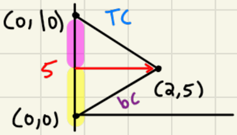
y-int = $(0,0)$

$y = mx + b = \frac{5}{2}x$

$$\int_0^2 \int_{\frac{5}{2}x}^{-\frac{5}{2}x + 10} f(x,y)\, dy\, dx$$

type one you are doing these with respect to x

type two you are doing these with respect to y



✱ going From left to right with respect to y

- notice here we now have two triangles and therefore we need to have two integrals

bc:

$y = \frac{5}{2}x$

We need to solve for x because we need this in terms of y

$x = \frac{2}{5}y$

TC:

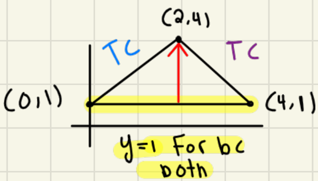
$y = -\frac{5}{2}x + 10$

We need to solve for x because we need it in terms of y

$x = -\frac{2}{5}y + 4$

$$\int_0^5 \int_0^{\frac{2}{5}y} F(x,y)\,dx\,dy + \int_5^{10} \int_0^{-\frac{2}{5}y+4} F(x,y)\,dx\,dy$$

we began from the bottom curve which begans at 0 and goes to 5 we the go to the second integral and start from where we left of and go to 10 we do this because we have two two triangles



type one

$0 \le x \le 4$

bottom $\le y \le$ top

TC:

$(0,1),\ (2,4)$
$x_1\, y_1 \quad x_2\, y_2$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4-1}{2-0} = \frac{3}{2}$

$y\text{-int} = (0,1)$

$y = mx + b$

$y = \frac{3}{2}x + 1$

TC:

$(4,1),\ (2,4)$
$x_1\, y_1 \quad x_2\, y_2$

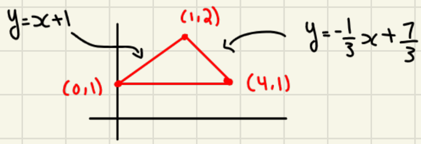
$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4-1}{2-4} = -\frac{3}{2}$

$y = mx + b = -\frac{3}{2}x + 7$

notice when you are dealing with the second half of the triangle its not intersecting on y-axis so plug $x=4$ into $y = \frac{3}{2}x + 1 = 7$

$$\int_0^2 \int_1^{3/2\,x+1} F(x,y)\,dy\,dx + \int_2^4 \int_1^{-\frac{3}{2}x+7} F(x,y)\,dy\,dx$$

Whatever variable is here you will have opposite here a triangle bounded by the Points: (0,1), (1,2), (4,1)



✱ if you are going to start with dy you are integrating Vertically if you began with dx you are integrating horizontally this also depends on how your triangle is Positioned but you might Need to break it up into two integrals

$$m=\frac{y_2-y_1}{x_2-x_1}=\frac{(2-1)}{(1-0)}=\frac{1}{1}=1 \qquad y=x+1$$

$$m=\frac{y_2-y_1}{x_2-x_1}=\frac{1-2}{4-1}=-\frac{1}{3}$$

• notice this does not have y-int

use:

$y-y_1=m(x-x_1)$

$y-2=-\frac{1}{3}(x-1)$

$y=-\frac{1}{3}x+\frac{7}{3}$

$$\int_0^1\int_{y=1}^{y=x+1} y^2\,dy\,dx+\int_1^4\int_{y=1}^{y=-1/3x+7/3} y^2\,dy\,dx$$

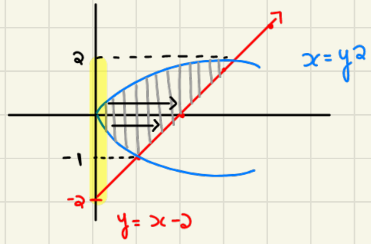
Starting with dy or going Vertical

or

$$\int_1^2\int_{x=y-1}^{x=-3y+7} y^2\,dx\,dy$$

Starting with dx or going horizontal

$\iint_D y\,dA$, D is bounded by $y = x-2$ and $x = y^2$



✯ if we bound the y-axis @ $-1 \leq y \leq 2$ and travel From the left to the right along the x-axis $y^2 \leq x \leq y+2$

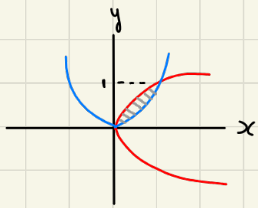
$$\int_{-1}^{2}\int_{x=y^2}^{x=y+2} y\,dx\,dy \Rightarrow \int_{y^2}^{y+2} y\,dx \Rightarrow \Big[yx\Big]_{y^2}^{y+2}$$

$$\Big[y(y+2) - y(y^2)\Big] \Rightarrow y^2+2y-y^3 = \int_{-1}^{2} y^2+2y-y^3\,dy$$

$$\left[\frac{y^3}{3} + y^2 - \frac{y^4}{4}\right]_{-1}^{2} \Rightarrow \left[\left(\frac{(2)^3}{3} + (2)^2 - \frac{(2)^4}{4}\right) - \left(\frac{(-1)^3}{3} + (-1)^2 - \frac{(-1)^4}{4}\right)\right]$$

$$\frac{8}{3} - \left(\frac{5}{12}\right) = \boxed{\frac{9}{4}}$$

under the Plane $3x+2y-z=0$ and above the region enclosed by the Parbolas $y=x^2$ and $x=y^2$



✯ we will set its bounds as $0 \leq y \leq 1$ along the y-axis and For the x-axis we will travel left to right

$3x+2y-z=0$

$z=3x+2y$

$$\int_0^1\int_{x=y^2}^{x=\sqrt{y}} 3x+2y\,dx\,dy$$

$$\int_0^1 \int_{x=y^2}^{x=\sqrt{y}} 3x+2y \; dx\,dy$$

$$\int_{y^2}^{\sqrt{y}} 3x+2y \; dx \Rightarrow \left[\frac{3x^2}{2}+2yx\right]_{y^2}^{\sqrt{y}}$$

$$\left[\left(\frac{3(\sqrt{y})^2}{2}+2y(\sqrt{y})\right)-\left(\frac{3(y^2)^2}{2}+2y(y^2)\right)\right]$$

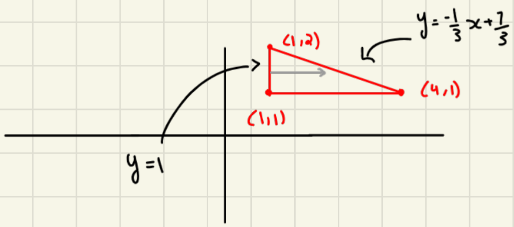
$$\frac{3y^2}{2}+2y\sqrt{y}-\frac{3y^4}{2}+2y^3$$

$$\int_0^1 \frac{3y^2}{2}+2y\sqrt{y}-\frac{3y^4}{2}+2y^3 \; dy$$

$$\left[\frac{3y^3}{6}+\frac{4}{5}y^{5/2}-\frac{3y^5}{10}+\frac{1}{2}y^4\right]_0^1$$

$$\frac{3}{6}+\frac{4}{5}-\frac{3}{10}+\frac{1}{2} = \boxed{\frac{3}{2}}$$

under the surface $z=xy$ and above the triangle with vertices $(1,1)$, $(4,1)$ and $(1,2)$



$$\int_1^4 \int_1^{y=-\frac{1}{3}x+\frac{7}{3}} xy \; dy\,dx$$

✗ this is Just set up and Not solved

we can bound the x @ $1 \le x \le 4$

$(1,2)$ $(4,1)$
$x_1\ y_1$ $x_2\ y_2$

$$m=\frac{y_2-y_1}{x_2-x_1}=\frac{1-2}{4-1}=\frac{-1}{3}$$

$$y-y_1=m(x-x_1)$$

$$y-2=\frac{-1}{3}(x-1) \Rightarrow y=-\frac{1}{3}x+\frac{7}{3}$$

Find the dimensions of the largest rectangular box with the largest volume if the total surface is given as 64 square inches

$V = xyz$ $\qquad$ $2xy + 2zx + 2zy = 64$ $\qquad$ $x > 0$

$y > 0$

$\nabla V = \langle yz, xz, xy \rangle$ $\qquad$ $\nabla g = 2\langle y+z, x+z, x+y \rangle$ $\qquad$ $z > 0$

$yz = 2\lambda(y+z)$ $\qquad$ $xz = 2\lambda(x+z)$ $\qquad$ $xy = 2\lambda(x+y)$

$\lambda = \dfrac{yz}{2(y+z)}$ $\qquad$ $\lambda = \dfrac{xz}{2(x+z)}$ $\qquad$ $\lambda = \dfrac{xy}{2(x+y)}$

$$\underbrace{\frac{yz}{2(y+z)} = \overbrace{\frac{xz}{2(x+z)}}^{} = \frac{xy}{2(x+y)}}_{}$$

$\dfrac{yz}{2(y+z)} = \dfrac{xz}{2(x+z)}$ $\qquad$ $\dfrac{xz}{2(x+z)} = \dfrac{yz}{2(y+x)}$

$2y\cancel{z}(x+z) = 2x\cancel{z}(y+z)$ $\qquad$ $2x\cancel{y}(y+x) = 2\cancel{y}x(x+z)$

$2\cancel{yx} + 2yz = 2\cancel{xy} + 2xz$ $\qquad$ $2\cancel{zy} + 2zx = 2yx + 2\cancel{yz}$

$\cancel{2}y\cancel{z} = \cancel{2}x\cancel{z}$ $\qquad$ $\cancel{2}z\cancel{x} = \cancel{2}y\cancel{x}$

$x = y$ $\qquad$ $z = y$

So: $x = y = z = a$

$2(xy + xz + yz) = 64$

$2(a^2 + a^2 + a^2) = 64$

$2(3a^2) = 64$

$6a^2 = 64 \rightarrow a^2 = \dfrac{32}{3}$

$a = \dfrac{4\sqrt{6}}{3}$

$$\boxed{\left(\frac{4\sqrt{6}}{3}, \frac{4\sqrt{6}}{3}, \frac{4\sqrt{6}}{3}\right)}$$